

Algorithm: Morphological Opening and Closing

Algorithm

Step 1: Start the program.

Step 2: Read the input image using OpenCV.

Step 3: Convert the input image from RGB/BGR to a grayscale image.

Step 4: Convert the grayscale image into a binary image using thresholding.

Step 5: Create a 5×5 rectangular structuring element (kernel).

Step 6: Perform morphological opening on the binary image. Opening is performed by erosion followed by dilation and is mainly used to remove small objects and noise.

Step 7: Perform morphological closing on the binary image. Closing is performed by dilation followed by erosion and is mainly used to fill small holes and gaps.

Step 8: Display the Original Image, Grayscale Image, Morphological Opening, and Morphological Closing.

Step 9: Stop the program.

Pseudocode

START

Read input image

Convert image to grayscale

Convert grayscale image to binary image

Create a 5×5 rectangular structuring element

Opening = Erosion(Binary Image)
 followed by Dilation

Closing = Dilation(Binary Image)
 followed by Erosion

Display:
 Original Image
 Grayscale Image
 Morphological Opening
 Morphological Closing

STOP

Result

Thus, the original image was converted into grayscale and binary form, and morphological opening and closing operations were successfully performed.